MERN STACK

DAY – 31

30 July 2026

PROJECT: PARKEASY - PARKING MANAGEMENT SYSTEM

LOGIN API DEVELOPMENT

• The Login API is developed to authenticate registered users in the ParkEasy parking management system.

• It verifies the email and password entered by the user with the data stored in the MongoDB database.

• The API is created using Express.js with a POST request.

API Endpoint:

```
POST /login
```

• The user email and password are received from the frontend using:

```
req.body
```

• The API extracts login credentials:

- Email
- Password

• The system searches for a user whose email and password match the database records using:

```
User.findOne({
email,
password
})
```

• If matching user details are found, authentication is successful.

• The server returns:

- Success status
- User details

• If the credentials are incorrect, the API returns: Invalid Email or Password

• Error handling is implemented using try-catch blocks to handle server-side issues.

Working

1. The app.post("/login") route creates the Login API.
2. User login credentials are received from the frontend.
3. The email and password are extracted from the request body.
4. The database is searched using the entered credentials.
5. If a matching user is found, login authentication is completed successfully.
6. User information is returned to the frontend.
7. The frontend uses the received user details for role-based navigation.
8. If no matching user is found, an error message is returned.

```
app.post("/login",async(req,res)=>{
    try{
        const {email,password}=req.body;

        const user=await User.findOne({
            email,
            password
        });

        if(!user)
            return res.json({
                success:false,
                message:"Invalid Email or Password"
            });

        res.json({
            success:true,
            user
        });

    }catch(err){
        res.json({
            success:false,
            message:err.message
        });
    }
});
```